Atmind Data Analytics Test 2026

Topic: Busy Buffet

Preface

We are managing a hotel breakfast buffet at Hotel Amber 85. We started promoting 3 things:

1. All you can eat
2. Price 159 weekday 199 weekend
3. 5 hour seating time

The promotion on tiktok had good response and we had a sudden increase in walk-in guests. The breakfast became very busy and difficult to manage so the data-team was sent to take a look and make sense of what to do next.

We sensed a variety of problems but we found it hard to decide what to do next.

The comments we get from our service staff were:

1. *"In-house (hotel) customers are unhappy that they have to wait for a table. Walk-in customers are also unhappy, when they queue up for a long time and leave the queue because they don't want to wait any longer".*
2. *"We are very busy every day of the week. If it's going to be this busy every week I think it's impossible to sustain this business. This buffet business is not possible for this hotel".*
3. *"Walk-in customers sit the whole day. It's very difficult to find seats for in-house customers. We don't have enough tables so when one customer sits for a long time it makes the queue very long".*

From all these comments the team decided to collect some data, the data fields and other relevant existing data guides can be found at the bottom of the file (Appendix1: Dataset Guide). There are also guidelines on how to read table setup data which is specifically created to solve table merging and splitting setups (Appendix2: Table Setup Guide).

Before data collection the team consulted with hotel management and found out that the actions that can be implemented are:

1. Reduce seating time (5 hours to less)
2. Increase price everyday to 259
3. Queue skipping for in-house guest

TASK

Using the dataset given and information given in the appendix complete the following tasks.

Task 1 (Mandatory):

For each staff comment, create visuals and analysis to ***prove if their statement is true or not***.

Task 2 (Mandatory):

For each of the recommended actions, create visual and analysis to ***disprove*** why each of them will **not** work.

Task 3 (Extra):

Pick 1 recommended action, create visual and analysis to support ***why it will work***. Describe and add through your own experience why you believe it to be the best solution. You can adjust specific details of the action and explain your reasoning. You can choose your preferred way of communicating as long as it is submittable through a google drive.

Tools and Submission Format:

Python Format:
Create and deploy a public streamlit dashboard to display relevant data and visuals. For commentary and analysis necessary in explaining the visual, you may screen shot each visual and write commentary through word doc or presentation. Feel free to reach out and discuss if you have other preferred methods of explaining your analysis. Thai and English are both accepted.

Appendix1: Dataset Guide

Table data in given file:
The given data is a replica of our real work data. It is messy and may have inaccuracies. However, a majority of the dataset will follow these guidelines:

1) service_no.

- Unique ID for each group on that day
- One group = one service number
- If a group uses multiple tables (like 2A, 2B), they still share the same service_no

2) pax

- Number of people in the group

3) queue_start

- Time the group arrives and takes a queue card
- If filled → the group waited
- If empty → the group did not wait (sat immediately)

4) queue_end

- Time the wait ends (either they get seated or leave)
- If both queue_start and queue_end exist → we can measure wait time
- If empty → treated as no queue

5) table_no.

- Table where the group sat
- Examples:
    - Single: 2, 7

    - Split: 2A, 7B
    - Combined: 13-14, 1A-1B
- Special case:
    - 99 = ate in queueing area

6) meal_start

- Time the group sits down
- Start of the meal

7) meal_end

- Time the group leaves
- End of the meal

8) Guest_type

- Type of guest:
    - In house : guest that stayed at the hotel
    - Walk in : guest that visit just for breakfast
- Used to compare guest source

Data Logic (How to Read the Data)

1) Walk-away

A group is a **walk-away** if:

- They waited in queue
- But never sat down

Simple rule:

- Queue exists + no meal → walk-away

2) Split and combined tables

We break tables into individual units:

- 13-14 → 13, 14
- 1A-1B → 1A, 1B

Not all table numbers are split.
Therefore we have some tables with ABC and some with none.

3. Edge Cases
1) No queue or no seating

- If only seating exists → no queue
- If only queue exists → no seating
- If nothing exists → no usable data

2) Quick definitions

- **Walk-away** = waited but never sat
- **Waited group** = has queue times
- **Direct seating** = sat without waiting]

Appendix2: Table Setup Guide

The restaurant is divided into Indoor and Outdoor seating areas.
Each table has a fixed name, and some tables are split into smaller sections (A, B, C).

Indoor Seating
Indoor tables are able to split:
1A, 1B
2A, 2B
3A, 3B
4A, 4B
5A, 5B
6A, 6B

Outdoor Seating
Outdoor tables include both split tables and full tables
<u>Split tables:</u>
7A, 7B, 7C
8A, 8B, 8C
9A, 9B, 9C
10A, 10B
11A, 11B
<u>Full tables (not split):</u>
12
13
14
15

Notes
Each split (A, B, C) is treated as a separate seating unit (basically 1 table or table for 2). When tables are combined (like 1A-1B), it means one group is using both sides (e.g. table for 4).

Table 99 (if used) is not part of indoor or outdoor seating; it is used when some customers just start getting food and eating in the queueing area.